

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview	1
2. Background & Business Objective	1
3. Product Purpose	2
4. System Functionalities	2
5. User Stories & Use Cases	3
6. Features Included (Scope Definition)	3
7. Features Not Included (Scope Control)	3
8. Assumptions & Constraints	4
9. Risks & Questions Throughout Development	4

Project Name: Grantly - SME Grant Hunter

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian SMEs face a highly fragmented, manual, and bureaucratic process when discovering and applying for government grants. They struggle to interpret strict eligibility requirements, prepare compliant documentation, and produce professional-grade proposals or pitch decks. This results in missed funding opportunities and inefficient application cycles.
- **Target Domain:** AI Systems & Agentic Workflow Automation for SME Financing and Government Grant Access.
- **Proposed Solution Summary:** Grantly is an AI-powered SME grant copilot that transforms fragmented business data into structured insights, evaluates eligibility against live grant databases, and autonomously generates submission-ready application packages. By leveraging a multi-agent architecture and DAG-based routing, Grantly shifts SMEs from manual preparation to intelligent, guided automation.

2. Background & Business Objective

- **Background of the Problem:** The traditional grant application workflow involves multiple disconnected steps: document preparation, eligibility interpretation, proposal writing, and submission formatting. These tasks require domain knowledge, time, and technical writing expertise, which many SMEs lack. Existing solutions are static and do not provide personalized, actionable guidance.

- **Importance of Solving This Issue:** Addressing this issue can significantly streamline how SMEs approach grant applications by cutting down preparation time from weeks to just minutes, while also increasing their chances of success through clearer guidance and structured support. It helps level the playing field by making funding opportunities more accessible to businesses that may lack prior experience or resources, and reduces the need to rely heavily on external consultants or intermediaries, allowing SMEs to manage the process more independently and efficiently.
- **Strategic Fit / Impact:** Grantly aligns with the global trend toward AI-native workflow automation, integrating structured AI reasoning into business processes. It demonstrates a scalable model for automating compliance-heavy workflows and supports Malaysia's digital economy initiatives by empowering SMEs.

3. Product Purpose

3.1. Main Goal of the System: To provide SMEs with an AI-driven copilot that simplifies grant discovery, evaluates readiness, and automates the creation of submission-ready application materials.

3.2. Intended Users (Target Audience):

- SME Owners (Primary Users)
- Startup Founders
- Entrepreneurs
- Business Development Managers
- Grant Consultants (Secondary Users)

4. System Functionalities

4.1. Description: Grantly operates as a cloud-based, AI-driven workflow system that ingests unstructured SME data, converts it into structured profiles, evaluates eligibility against grant requirements, and routes users through personalized pathways using a Directed Acyclic Graph (DAG) architecture.

4.2. Key Functionalities:

- **Progressive Onboarding & Gated Dashboard:** Users are allowed to access the main dashboard immediately after logging in, where key features are visible but intentionally locked to encourage engagement. This approach reduces initial friction while guiding users to complete their SME profile in order to unlock personalized grant insights and system capabilities.
- **Conversational Editing:** [e.g. Facilitates continuous modification of the existing design by speaking to canvas itself/ texting.]

- **Tri-Fold Profile Builder:** The system provides three flexible methods for users to input their data: uploading documents such as PDFs or certificates, completing a guided questionnaire with structured inputs, or manually entering information through text fields, ensuring accessibility for users with different levels of readiness and data availability.
- **AI Extractor Agent:** The Extractor Agent processes both uploaded documents and user inputs by applying OCR and text parsing techniques, converting unstructured data into a standardized SMEProfile JSON format while also identifying any missing mandatory documents required for evaluation.
- **Grant Matching & Evaluator Agent:** The Evaluator Agent compares the structured SME profile against a database of grant requirements, performing logical and mathematical checks to determine eligibility, and produces outputs such as a readiness score, an evidence trace for transparency, and an explainable match score for user understanding.
- **DAG-Based Decision Routing:** The system uses a Directed Acyclic Graph (DAG) routing mechanism to guide users based on their readiness score, automatically directing those below 80% to a guidance pathway and those above 80% to an automated document generation workflow.
- **Coach Agent (Guidance Mode):** For users who are not yet eligible, the Coach Agent generates a clear and actionable roadmap that explains missing requirements in simple terms and provides step-by-step instructions to help users meet the necessary criteria.
- **Drafter Agent (Automation Mode):** The Drafter Agent supports two modes: an evaluation mode that analyzes and critiques existing pitch decks, and a generation mode that creates a complete set of documents including a business proposal, presentation script, and structured pitch deck content.

- **Export Pipeline:** The system converts the structured outputs into professional file formats using backend tools, generating PDF, DOCX, and PPTX documents, which are then compiled into a single downloadable ZIP file containing all submission-ready materials.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

Grantly uses Z.ai GLM API due to its strong structured output capabilities, JSON compliance, and suitability for multi-agent workflows requiring deterministic responses.

4.3.2. Prompting Strategy:

The system adopts a multi-step agentic prompting strategy, where tasks are broken down into smaller, specialized steps handled by different agents instead of relying on a single prompt. This approach is combined with a divide-and-conquer method, where large generation tasks are split into parallel prompts and executed asynchronously to improve efficiency and avoid latency issues. Additionally, schema-driven prompting is enforced by embedding strict JSON structures within prompts to ensure outputs are consistent, structured, and easy to validate. This strategy fits the use case well because the system requires high accuracy, explainability, and reliable data processing across multiple stages such as extraction, evaluation, and document generation, making a structured and modular prompting approach more effective than simple zero-shot or few-shot methods.

4.3.3. Context & Input Handling:

The system is designed to handle unstructured inputs such as PDFs and mixed-form data by first applying OCR and text extraction before passing the content into the processing pipeline. For large inputs, the system uses chunking to break documents into smaller segments so they can be processed without exceeding model limits, while truncation is applied when only the most relevant sections are needed. To maintain performance and control costs, a maximum file size threshold is enforced at the backend example like around 10MB for uploads, and any inputs that

exceed this limit are automatically rejected before reaching the AI processing stage, with clear feedback provided to the user to re-upload a compliant file.

4.3.4. Fallback & Failure Behavior:

When the model produces off-topic, unusable, or potentially hallucinated responses, the system enforces strict JSON schema validation to ensure outputs conform to predefined structures, rejecting any malformed or irrelevant results before they reach the user. If an API failure or invalid response occurs, an automatic retry mechanism with exponential backoff is triggered to recover from transient issues. In cases where retries still fail, the system falls back to a deterministic, rule-based logic to maintain basic functionality, such as displaying grant requirements without AI-generated insights. Throughout this process, users are presented with a clear and graceful error message, ensuring transparency while maintaining system reliability without exposing raw or misleading AI outputs.

5. User Stories & Use Cases

- **User Stories:** As an SME owner, I want to seamlessly upload or input my business data in any format and instantly receive a structured profile, a clear and explainable readiness score, and personalized guidance on missing requirements, so that I can quickly understand my eligibility and confidently generate a complete, professional-grade grant application without relying on external consultants or spending excessive time on manual preparation.
- **Use Cases (Main Interactions):**

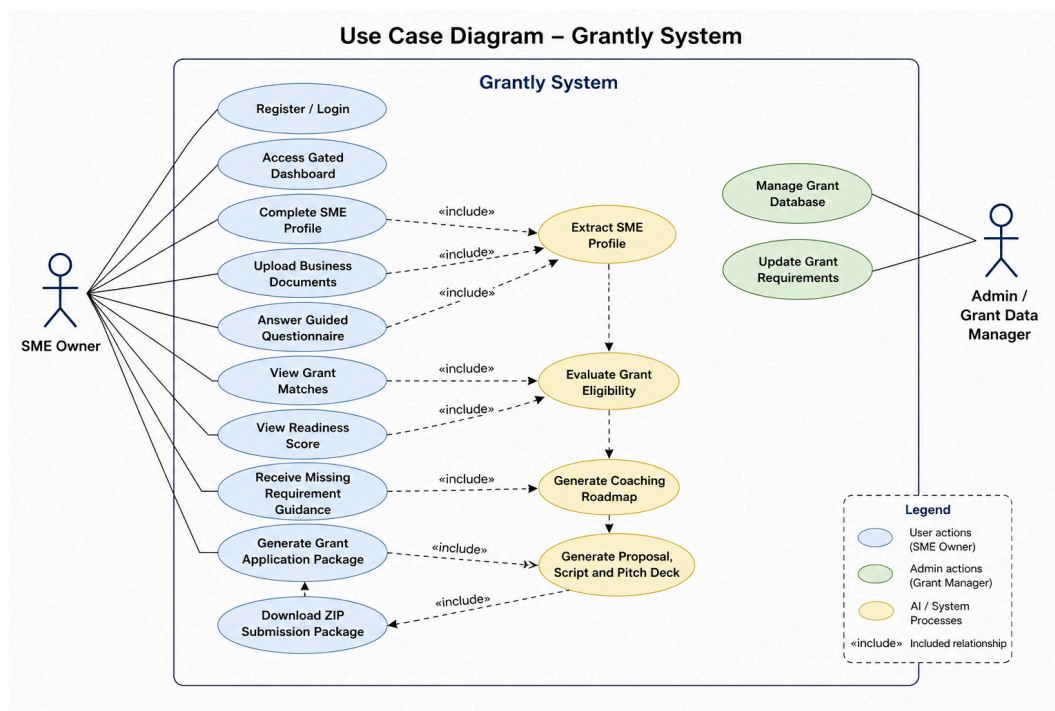
Onboarding Flow: The onboarding flow begins when the user logs into the system and is immediately directed to the main dashboard, where core features are visible but locked. The user is then guided to complete their SME profile through various input methods, and once sufficient information is provided, the system unlocks personalized functionalities such as grant matching and readiness evaluation.

Grant Matching: After the SME profile is completed, the system processes the structured data and compares it against a database of available grants. The user is then presented with a list of relevant grants along with a readiness score, explainable matching insights, and a visual roadmap outlining the required steps for application.

DAG Routing: Based on the evaluated readiness score, the system automatically routes the user through a decision pathway using a DAG structure. Users who are not yet eligible are directed to a guidance mode with actionable recommendations, while those who meet the criteria are routed to the automation pathway for document generation.

Application Generation: Once a user selects a specific grant and meets the required criteria, the system generates a complete set of application materials, including a business proposal, presentation script, and pitch deck content. These outputs are then converted into standard file formats and compiled into a downloadable ZIP package for submission.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- A hybrid onboarding system that allows users to input data through document uploads, guided forms, or manual entry.
- An AI-powered extraction engine that converts unstructured SME data into a structured and standardized profile.
- An explainable grant matching system that evaluates eligibility and provides transparent readiness scores with supporting evidence.
- A DAG-based decision routing mechanism that dynamically guides users to either guidance or automation workflows based on their readiness level.
- An AI-driven document generation system that produces business proposals, presentation scripts, and pitch deck content.
- A one-click export pipeline that converts generated outputs into submission-ready files in multiple formats (PDF, DOCX, PPTX).

7. Features Not Included (Scope Control)

- Direct submission to government portals
- Real-time API integration with SSM or MDEC
- Multi-user collaboration features
- Full production-level scalability (MVP limitation)

8. Assumptions & Constraints

- **LLM Cost Constraint:** The system is designed to manage token usage efficiently, with an estimated moderate cost per user session due to multiple AI interactions across extraction, evaluation, and drafting stages. To control inference costs at scale, the system adopts structured JSON prompting, avoids unnecessary long context inputs, and uses a divide-and-conquer strategy with parallel prompts, ensuring that only essential text processing is handled by the model while file generation is offloaded to backend services.

- **Technical Constraints:** The system is currently limited to generating outputs in standard document formats such as PDF, DOCX, and PPTX, and does not support direct integration with external government APIs or advanced frameworks beyond its current stack. Additionally, the platform relies on third-party AI services, which introduces dependency on API availability and response consistency.
- **Performance Constraints:** Due to the complexity of generating high-quality structured outputs such as proposals and pitch decks, the system imposes limits on input size and generation length to prevent excessive latency and API timeouts. As a result, large tasks are split into smaller parallel processes, and there may be slight delays during peak processing stages.
- **User Input:** The system depends on users to provide accurate and sufficient business data, either through uploads or manual input, as the AI cannot generate meaningful outputs without proper context. Incomplete or low-quality inputs may affect the accuracy of evaluation and generated results, and users are required to review and validate outputs before final submission.

9. Risks & Questions Throughout Development

- **API Timeouts & System Reliability:**

The system faces potential risks such as API timeouts (e.g., 504 Gateway errors), LLM hallucination, and limitations in handling large or unstructured inputs. These risks are mitigated through strategies such as asynchronous parallel prompting to reduce latency, strict JSON schema validation to prevent hallucinated outputs, and input size control with chunking and file rejection mechanisms to ensure stable processing.

Question: How can the system ensure consistent performance and prevent failures when the AI service experiences latency or timeouts?

- **User Drop-off During Onboarding:**

The system uses a progressive onboarding approach with a gated dashboard that immediately shows value while encouraging completion. By revealing locked features and offering multiple input methods (upload, guided, manual), the system reduces friction and motivates users to complete their profiles.

Question: How can the system ensure users complete onboarding despite the need to provide detailed business information?

- **LLM Hallucination & Output Accuracy:**

The system enforces strict JSON schema validation and structured prompting to constrain outputs within predefined formats, ensuring consistency and reducing hallucination risks. Furthermore, the AI is restricted to only use user-provided data, and critical evaluations (e.g., financial thresholds) are validated through deterministic backend logic before being presented to users.

Question: How can the system guarantee that AI-generated outputs remain accurate and do not introduce misleading or fabricated information?